

Introduction to Electronic Design Automation

Homework 4

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1 Partition

Initial partition

$$A = \{a, b, c\}, \quad B = \{d, e, f\}$$

$$\text{Initial cut cost} = (3 + 3 + 2) + (2 + 4 + 1) + (0 + 2 + 2) = 19$$

Iteration 1

$$A = \{a, b, c\}, \quad B = \{d, e, f\}$$

$$\text{Current cut cost} = 19$$

Step 1

Current I , E , and D values:

Vertex	I	E	$D = E - I$	Locked
a	$2 + 1 = 3$	$3 + 3 + 2 = 8$	$8 - 3 = 5$	no
b	$2 + 0 = 2$	$2 + 4 + 1 = 7$	$7 - 2 = 5$	no
c	$1 + 0 = 1$	$0 + 2 + 2 = 4$	$4 - 1 = 3$	no
d	$1 + 0 = 1$	$3 + 2 + 0 = 5$	$5 - 1 = 4$	no
e	$1 + 1 = 2$	$3 + 4 + 2 = 9$	$9 - 2 = 7$	no
f	$0 + 1 = 1$	$2 + 1 + 2 = 5$	$5 - 1 = 4$	no

Candidate gains:

$$g_{ad} = D_a + D_d - 2c_{ad} = 5 + 4 - 2 \cdot 3 = 3$$

$$g_{ae} = D_a + D_e - 2c_{ae} = 5 + 7 - 2 \cdot 3 = 6$$

$$g_{af} = D_a + D_f - 2c_{af} = 5 + 4 - 2 \cdot 2 = 5$$

$$g_{bd} = D_b + D_d - 2c_{bd} = 5 + 4 - 2 \cdot 2 = 5$$

$$g_{be} = D_b + D_e - 2c_{be} = 5 + 7 - 2 \cdot 4 = 4$$

$$g_{bf} = D_b + D_f - 2c_{bf} = 5 + 4 - 2 \cdot 1 = 7$$

$$g_{cd} = D_c + D_d - 2c_{cd} = 3 + 4 - 2 \cdot 0 = 7$$

$$g_{ce} = D_c + D_e - 2c_{ce} = 3 + 7 - 2 \cdot 2 = 6$$

$$g_{cf} = D_c + D_f - 2c_{cf} = 3 + 4 - 2 \cdot 2 = 3$$

Therefore, the maximum gain is

$$\hat{g}_1 = g_{bf} = 7$$

Temporarily swap b and f , and lock them.

Step 2

Current I , E , and D values:

Vertex	I	E	$D = E - I$	Locked
a	$1 + 2 = 3$	$2 + 3 + 3 = 8$	$8 - 3 = 5$	no
b	$2 + 4 = 6$	$2 + 0 + 1 = 3$	$3 - 6 = -3$	yes
c	$1 + 2 = 3$	$0 + 0 + 2 = 2$	$2 - 3 = -1$	no
d	$2 + 1 = 3$	$3 + 0 + 0 = 3$	$3 - 3 = 0$	no
e	$4 + 1 = 5$	$3 + 2 + 1 = 6$	$6 - 5 = 1$	no
f	$2 + 2 = 4$	$1 + 0 + 1 = 2$	$2 - 4 = -2$	yes

Candidate gains:

$$g_{ad} = D_a + D_d - 2c_{ad} = 5 + 0 - 2 \cdot 3 = -1$$

$$g_{ae} = D_a + D_e - 2c_{ae} = 5 + 1 - 2 \cdot 3 = 0$$

$$g_{cd} = D_c + D_d - 2c_{cd} = -1 + 0 - 2 \cdot 0 = -1$$

$$g_{ce} = D_c + D_e - 2c_{ce} = -1 + 1 - 2 \cdot 2 = -4$$

Therefore, the maximum gain is

$$\hat{g}_2 = g_{ae} = 0$$

Temporarily swap a and e , and lock them.

Step 3

Current I , E , and D values:

Vertex	I	E	$D = E - I$	Locked
a	$2 + 3 = 5$	$1 + 3 + 2 = 6$	$6 - 5 = 1$	yes
b	$2 + 2 = 4$	$0 + 4 + 1 = 5$	$5 - 4 = 1$	yes
c	$2 + 2 = 4$	$1 + 0 + 0 = 1$	$1 - 4 = -3$	no
d	$3 + 2 = 5$	$0 + 1 + 0 = 1$	$1 - 5 = -4$	no
e	$2 + 1 = 3$	$3 + 4 + 1 = 8$	$8 - 3 = 5$	yes
f	$2 + 1 = 3$	$2 + 1 + 0 = 3$	$3 - 3 = 0$	yes

Candidate gains:

$$g_{cd} = D_c + D_d - 2c_{cd} = -3 + -4 - 2 \cdot 0 = -7$$

Therefore, the maximum gain is

$$\hat{g}_3 = g_{cd} = -7$$

Temporarily swap c and d , and lock them.

Summary of Iteration 1

$$\hat{g}_1 = g_{bf} = 7,$$

$$\hat{g}_2 = g_{ae} = 0,$$

$$\hat{g}_3 = g_{cd} = -7$$

The partial sums are:

$$G_1 = \sum_{j=1}^1 \hat{g}_j = 7,$$

$$G_2 = \sum_{j=1}^2 \hat{g}_j = 7,$$

$$G_3 = \sum_{j=1}^3 \hat{g}_j = 0$$

Thus,

$$\max_k G_k = 7 \quad \text{at} \quad k = 1$$

Since the largest partial sum is positive, apply the first 1 swap(s):

- Swap b and f .

After applying the selected swap(s),

$$A = \{a, c, f\}, \quad B = \{b, d, e\}$$

$$\text{New cut cost} = 12$$

Iteration 2**Step 1**

Current I , E , and D values:

Vertex	I	E	$D = E - I$	Locked
a	$1 + 2 = 3$	$2 + 3 + 3 = 8$	$8 - 3 = 5$	no
b	$2 + 4 = 6$	$2 + 0 + 1 = 3$	$3 - 6 = -3$	no
c	$1 + 2 = 3$	$0 + 0 + 2 = 2$	$2 - 3 = -1$	no
d	$2 + 1 = 3$	$3 + 0 + 0 = 3$	$3 - 3 = 0$	no
e	$4 + 1 = 5$	$3 + 2 + 1 = 6$	$6 - 5 = 1$	no
f	$2 + 2 = 4$	$1 + 0 + 1 = 2$	$2 - 4 = -2$	no

Candidate gains:

$$g_{ab} = D_a + D_b - 2c_{ab} = 5 + -3 - 2 \cdot 2 = -2$$

$$g_{ad} = D_a + D_d - 2c_{ad} = 5 + 0 - 2 \cdot 3 = -1$$

$$g_{ae} = D_a + D_e - 2c_{ae} = 5 + 1 - 2 \cdot 3 = 0$$

$$g_{cb} = D_c + D_b - 2c_{cb} = -1 + -3 - 2 \cdot 0 = -4$$

$$g_{cd} = D_c + D_d - 2c_{cd} = -1 + 0 - 2 \cdot 0 = -1$$

$$g_{ce} = D_c + D_e - 2c_{ce} = -1 + 1 - 2 \cdot 2 = -4$$

$$g_{fb} = D_f + D_b - 2c_{fb} = -2 + -3 - 2 \cdot 1 = -7$$

$$g_{fd} = D_f + D_d - 2c_{fd} = -2 + 0 - 2 \cdot 0 = -2$$

$$g_{fe} = D_f + D_e - 2c_{fe} = -2 + 1 - 2 \cdot 1 = -3$$

Therefore, the maximum gain is

$$\hat{g}_1 = g_{ae} = 0$$

Temporarily swap a and e , and lock them.

Step 2

Current I , E , and D values:

Vertex	I	E	$D = E - I$	Locked
a	$2 + 3 = 5$	$1 + 3 + 2 = 6$	$6 - 5 = 1$	yes
b	$2 + 2 = 4$	$0 + 4 + 1 = 5$	$5 - 4 = 1$	no
c	$2 + 2 = 4$	$1 + 0 + 0 = 1$	$1 - 4 = -3$	no
d	$3 + 2 = 5$	$0 + 1 + 0 = 1$	$1 - 5 = -4$	no
e	$2 + 1 = 3$	$3 + 4 + 1 = 8$	$8 - 3 = 5$	yes
f	$2 + 1 = 3$	$2 + 1 + 0 = 3$	$3 - 3 = 0$	no

Candidate gains:

$$g_{cb} = D_c + D_b - 2c_{cb} = -3 + 1 - 2 \cdot 0 = -2$$

$$g_{cd} = D_c + D_d - 2c_{cd} = -3 + -4 - 2 \cdot 0 = -7$$

$$g_{fb} = D_f + D_b - 2c_{fb} = 0 + 1 - 2 \cdot 1 = -1$$

$$g_{fd} = D_f + D_d - 2c_{fd} = 0 + -4 - 2 \cdot 0 = -4$$

Therefore, the maximum gain is

$$\hat{g}_2 = g_{fb} = -1$$

Temporarily swap f and b , and lock them.

Step 3

Current I , E , and D values:

Vertex	I	E	$D = E - I$	Locked
a	$3 + 2 = 5$	$2 + 1 + 3 = 6$	$6 - 5 = 1$	yes
b	$0 + 4 = 4$	$2 + 2 + 1 = 5$	$5 - 4 = 1$	yes
c	$0 + 2 = 2$	$1 + 0 + 2 = 3$	$3 - 2 = 1$	no
d	$3 + 0 = 3$	$2 + 0 + 1 = 3$	$3 - 3 = 0$	no
e	$4 + 2 = 6$	$3 + 1 + 1 = 5$	$5 - 6 = -1$	yes
f	$2 + 0 = 2$	$1 + 2 + 1 = 4$	$4 - 2 = 2$	yes

Candidate gains:

$$g_{cd} = D_c + D_d - 2c_{cd} = 1 + 0 - 2 \cdot 0 = 1$$

Therefore, the maximum gain is

$$\hat{g}_3 = g_{cd} = 1$$

Temporarily swap c and d , and lock them.

Summary of Iteration 2

$$\hat{g}_1 = g_{ae} = 0,$$

$$\hat{g}_2 = g_{fb} = -1,$$

$$\hat{g}_3 = g_{cd} = 1$$

The partial sums are:

$$G_1 = \sum_{j=1}^1 \hat{g}_j = 0,$$

$$G_2 = \sum_{j=1}^2 \hat{g}_j = -1,$$

$$G_3 = \sum_{j=1}^3 \hat{g}_j = 0$$

Thus,

$$\max_k G_k = 0 \quad \text{at} \quad k = 1$$

Since the largest partial sum is not positive, the algorithm stops.

Final result

$$A = \{a, c, f\}, \quad B = \{b, d, e\}$$

$$\text{Final cut cost} = 12$$

2 Floorplanning

3 Wire-length Estimation

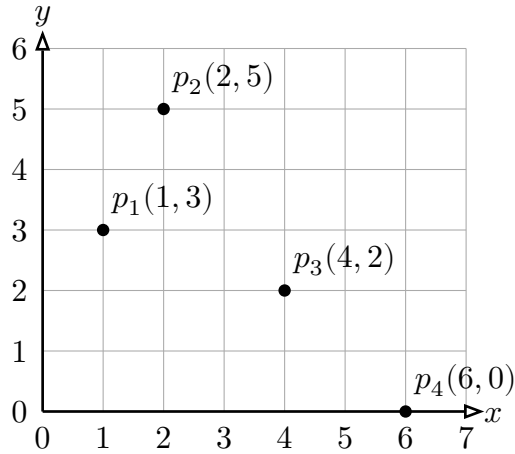


Figure 1: Pins of net N on a grid layout surface.

3. (a) Semi-perimeter Length

The semi-perimeter length is $(6 - 1) + (5 - 0) = 5 + 5 = 10$

3. (b) Complete Graph Length

$$\text{wirelength} \approx \frac{2}{n} \sum_{i,j \in \text{net}} \text{dist}(i,j) = \frac{2}{4}(3 + 4 + 8 + 5 + 9 + 4) = 16.5$$

3. (c) Minimum Spanning Tree Length

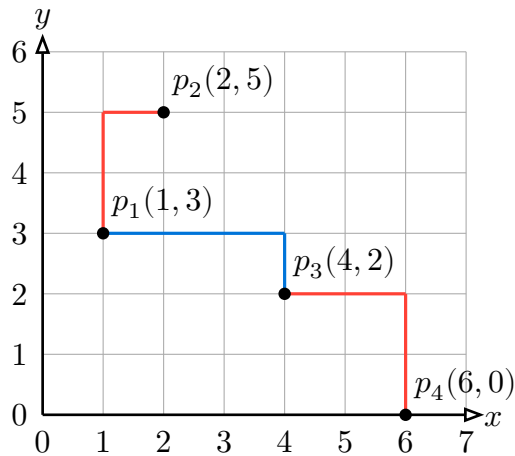


Figure 2: The Minimum Spanning Tree.

The length of the minimum spanning tree is $3 + 4 + 4 = 11$.

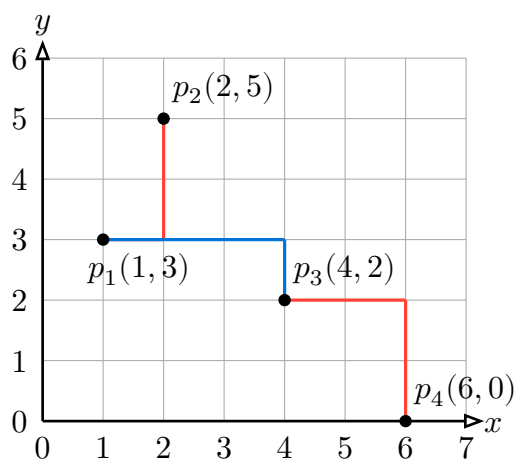
3. (d) Minimum Steiner Tree Length

Figure 3: The Minimum Steiner Tree.

The length of the minimum Steiner tree is $2 + 4 + 4 = 10$.

4 Force-Directed Placement

5 Maze Routing

6 Channel Routing

7 Testing